

Unit 9, Lesson 4: Reflections on the Coordinate Plane

Benchmark

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

Learning Targets

- ☐ I can perform a reflection on the coordinate plane.
- ☐ I can describe relationships between figures after a reflection.

9.4.1 Warm-Up: Think Like a Mathematician

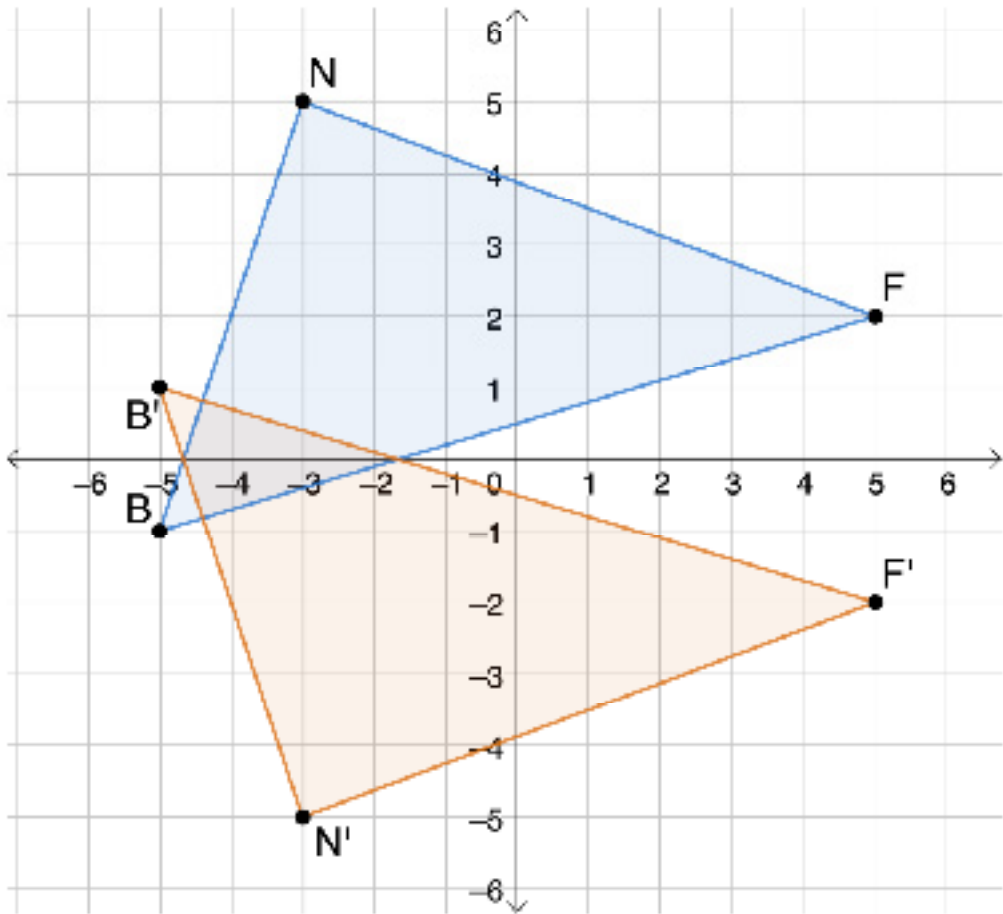
1. The image shows the beginning of a bathroom remodel.



List some of the mathematics that may be needed to complete a remodel such as this one. Include at least three things on your list.

9.4.2 Guided Instruction: Reflections on the Coordinate Plane

1. On the coordinate plane shown, $\triangle NBF$ is reflected to create $\triangle N'B'F'$.

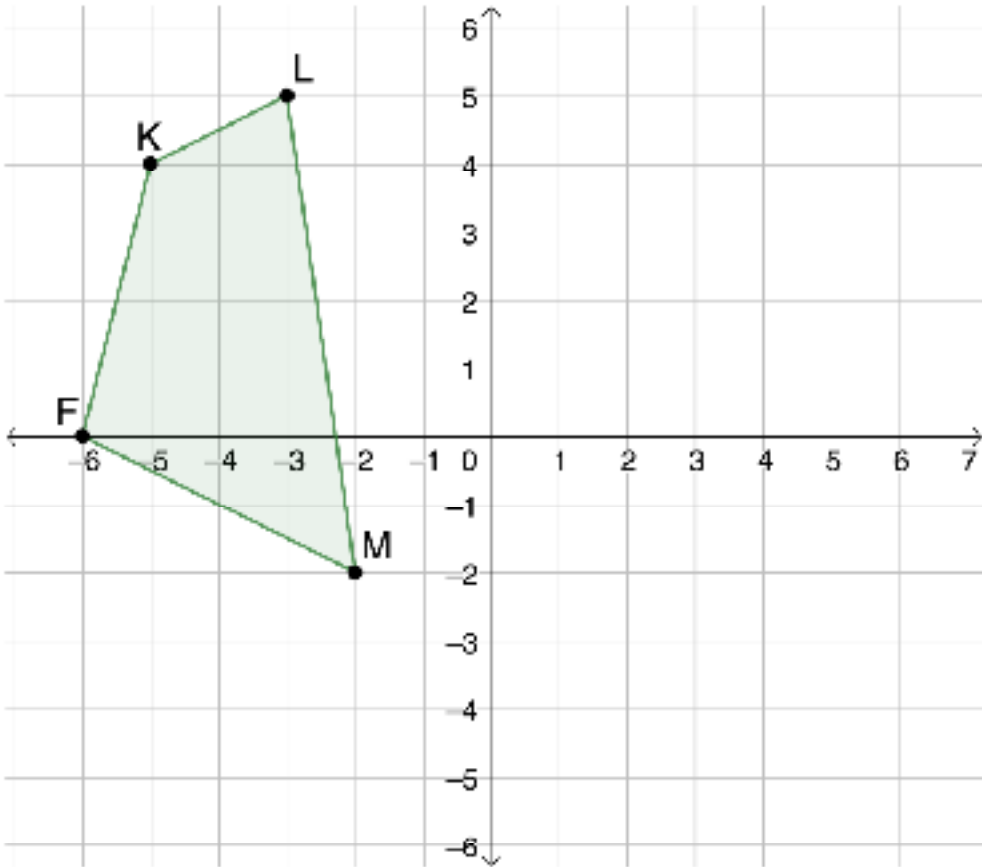


- a. What is the line of reflection?
- b. Complete the table with the coordinates of each vertex.

Vertex	Coordinates	Vertex	Coordinates
Point N		Point N'	
Point B		Point B'	
Point F		Point F'	

- c. What do you notice about the coordinates of the corresponding vertices in the preimage and the image?

2. Quadrilateral $FKLM$ is shown on the coordinate grid.



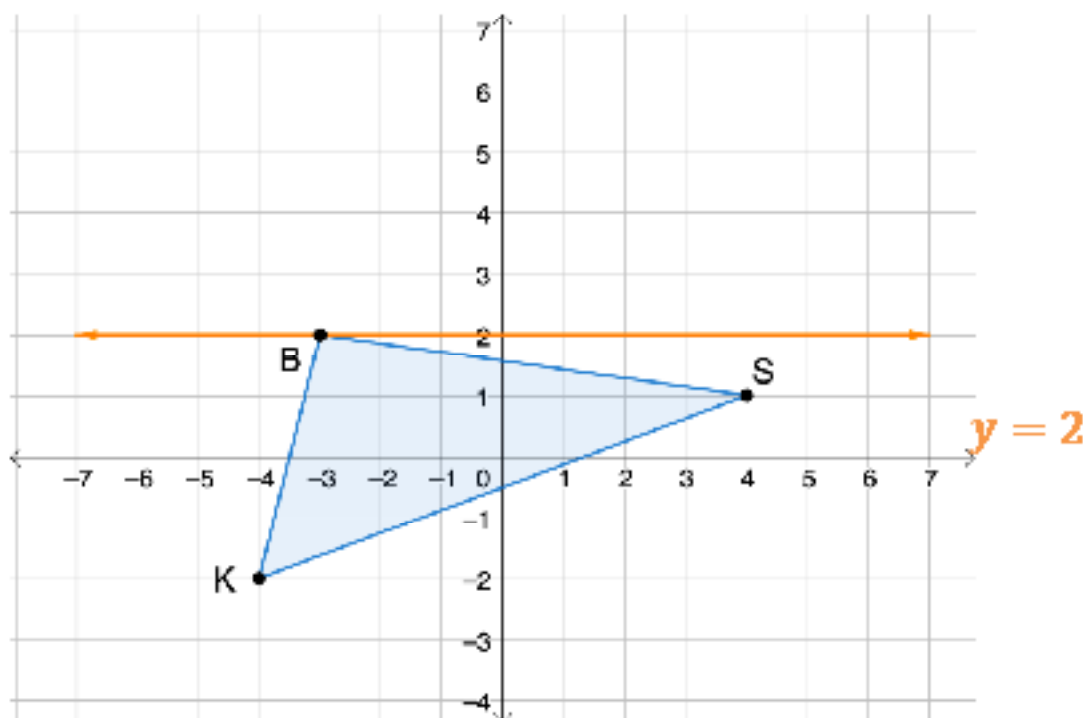
- a. Complete the left side of the table for points F , K , L , and M .

Vertex	Distance From y -axis	Coordinates	Vertex	Distance From y -axis	Coordinates
F			F'		
K			K'		
L			L'		
M			M'		

- b. Reflect quadrilateral $FKLM$ across the y -axis. Label the image quadrilateral $F'K'L'M'$.
- c. Complete the right side of the table in Part A for vertices F' , K' , L' , and M' .
- d. Describe what you notice about the coordinates of the corresponding vertices in the preimage and the image.

3. Figures can be reflected across any line on the coordinate plane, not only the x - or y -axis.

Triangle SBK and the line $y = 2$ are shown on the coordinate grid.



- a. Reflect $\triangle SBK$ across the line $y = 2$. Label the image $\triangle S'B'K'$.

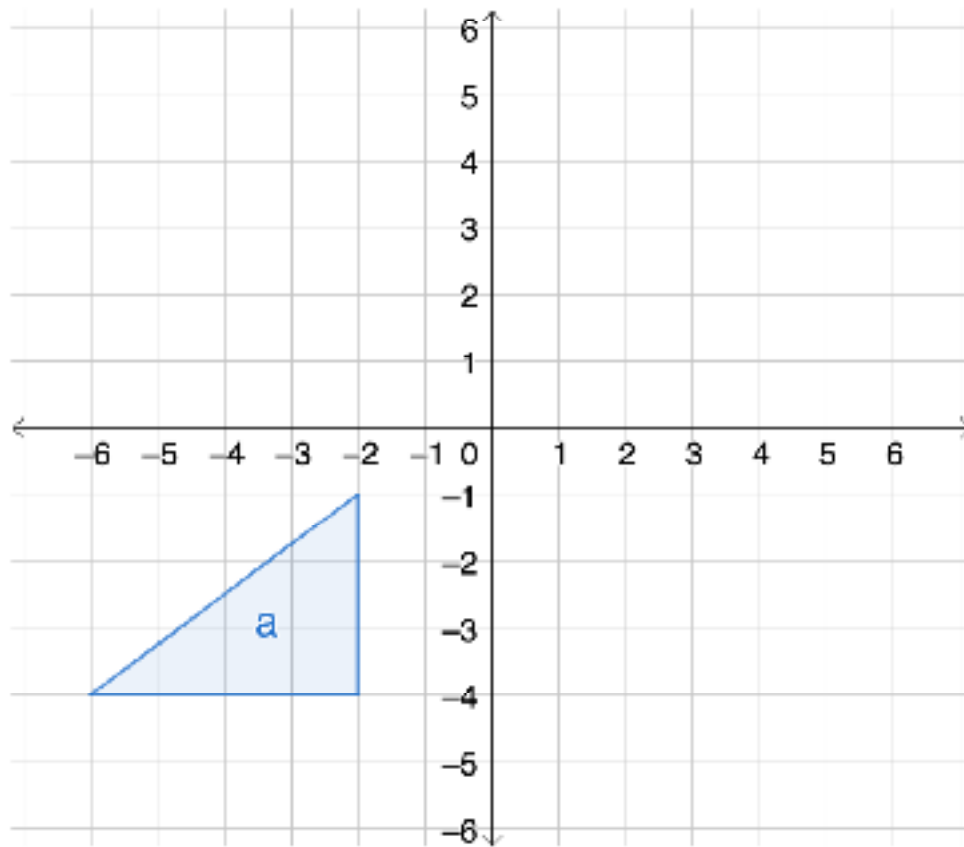
b. Complete the table with the coordinates of the vertices of the preimage and image.

Vertex	Coordinates	Vertex	Coordinates
Point <i>S</i>		Point <i>S'</i>	
Point <i>B</i>		Point <i>B'</i>	
Point <i>K</i>		Point <i>K'</i>	

c. Describe the relationship between the coordinates of points *B* and *B'*.

9.4.3 Collaboration: Which Quadrant?

1. A triangle is shown on the coordinate plane.



- a. Without performing the transformation, discuss with your partner what the image of the triangle would look like after a reflection across the y -axis.
- b. Complete the statement.

The preimage of the triangle is in quadrant

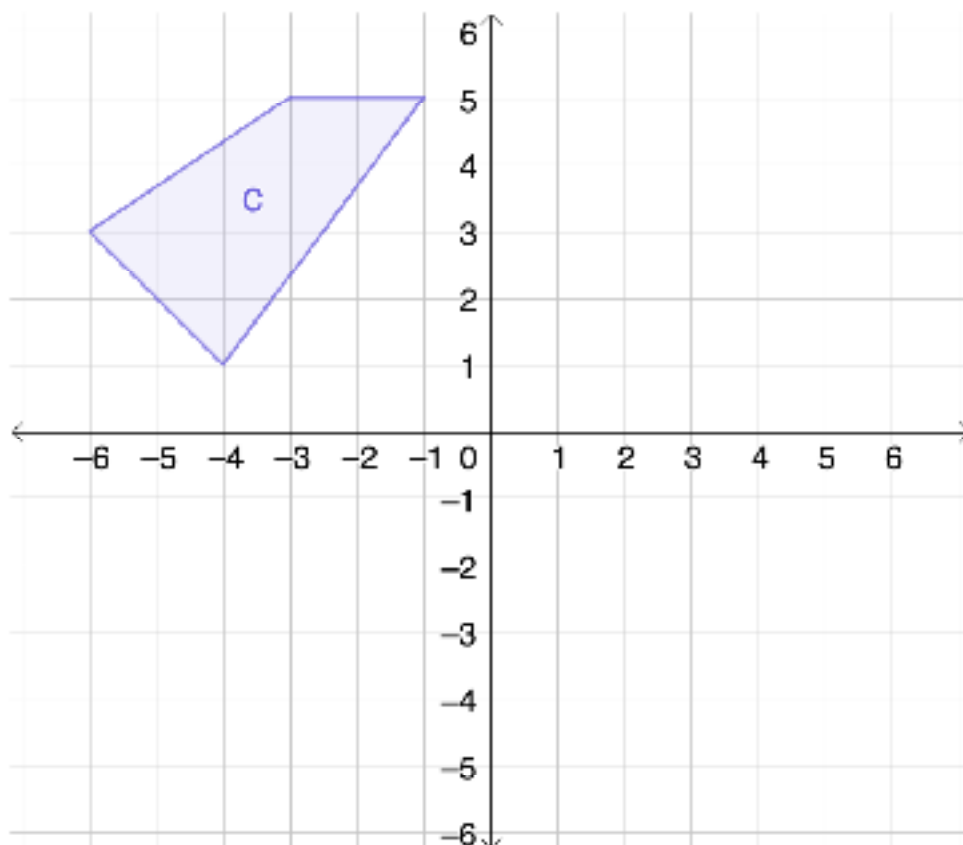
- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

The image of the triangle after a reflection across

the y -axis would lie in quadrant

- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

2. A quadrilateral is shown on the coordinate plane.



Without performing the transformation, complete the statement.

The preimage of the quadrilateral is in quadrant

- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

The image of the quadrilateral after a reflection

across the x -axis would lie in quadrant

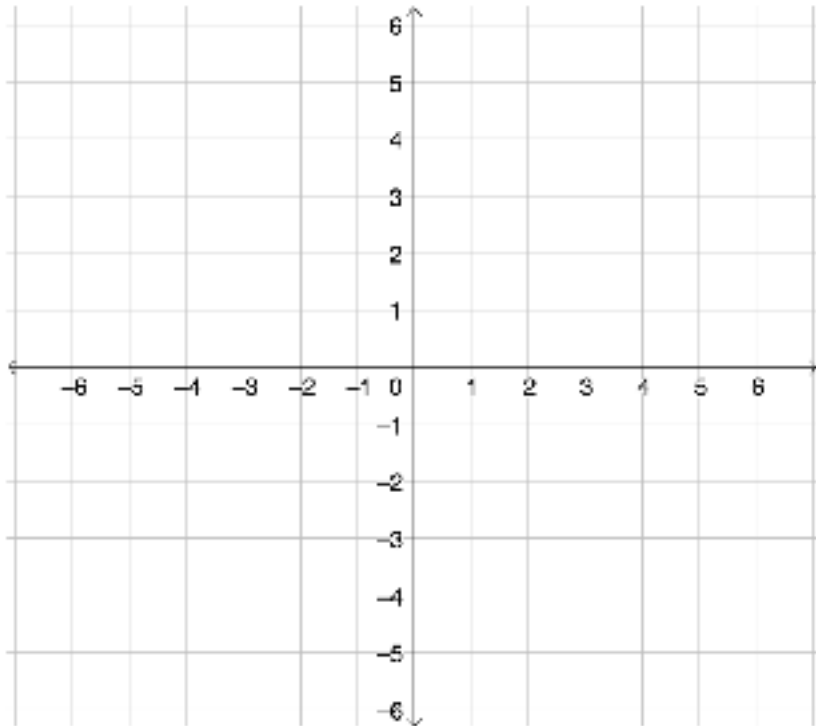
- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

9.4.4 Your Turn!

1. The vertices of $\triangle KMR$ have their coordinates given in the table.

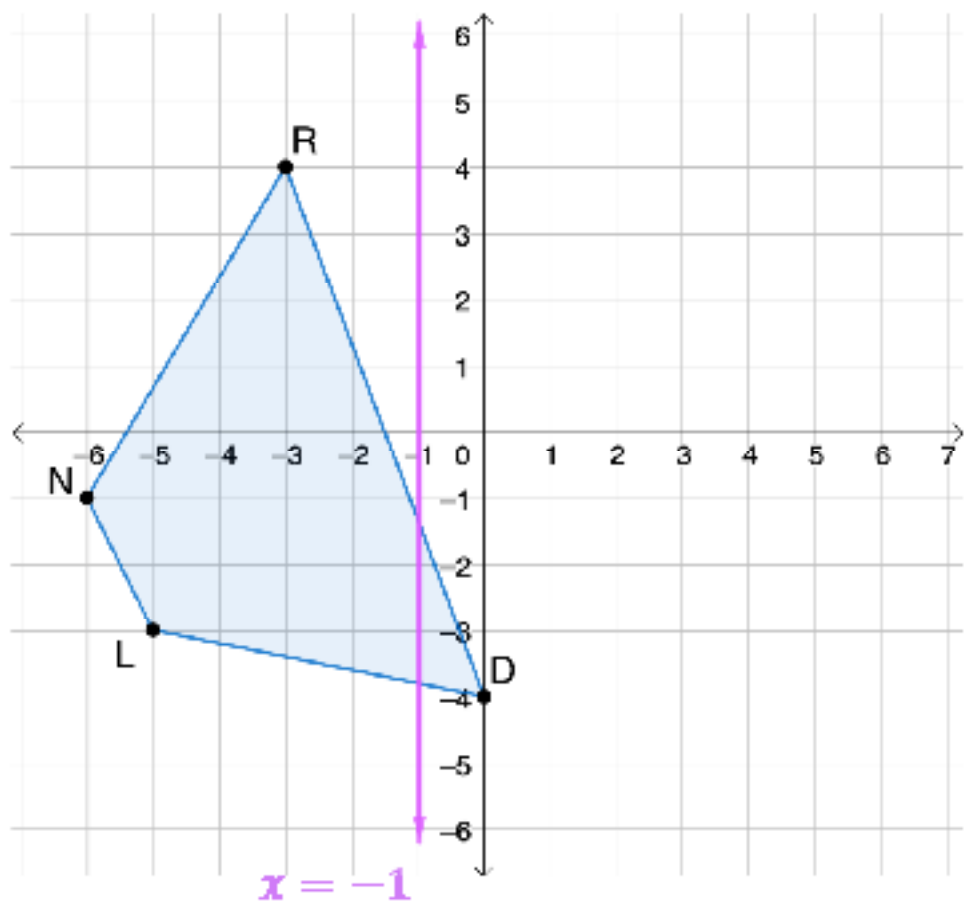
Vertex	Coordinates	Vertex	Coordinates
Point K	$(-3, -5)$	Point K'	
Point M	$(6, -2)$	Point M'	
Point R	$(-5, -1)$	Point R'	

a. Plot $\triangle KMR$ on the coordinate grid.



- b. Reflect $\triangle KMR$ across the x -axis. Label the image $\triangle K'M'R'$.
- c. Complete the table above for the image.
- d. Describe the relationship between the corresponding vertices of the preimage and image.

2. Quadrilateral $RD LN$ and line $x = -1$ are shown on the coordinate grid.



- a. Reflect quadrilateral $RD LN$ over the line of reflection, $x = -1$. Label the image $R'D'L'N'$.
- b. Describe how each pair of corresponding vertices relate to each other compared to the line of reflection.

Corresponding Vertices	Description of Relationship
R and R'	
D and D'	
L and L'	
N and N'	

Complete each statement as you reflect on today's learning.

- Unit 9 – Lesson 4 | 53

